

# Probing for Vietnamese Tone Representations in Whisper and PhoWhisper

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## Abstract

Vietnamese is a tonal language with five to six (depending on the dialect) lexical tones. This poses a challenge for speech recognition systems due to fine-grained pitch information that may be missed by standard systems, especially those trained on non-tonal languages. This paper explores whether OpenAI’s automatic speech recognition system, Whisper, is able to encode some internal representation of Vietnamese lexical tones. Using linear probing, we test how much tonal information is able to be encoded within the model’s hidden layers and whether or not it can accurately identify tones. The findings show that Whisper was showing improvement at later layers, but a Vietnamese fine-tuned model, PhoWhisper, showed even more improvement near the middle layers of the model. PhoWhisper had a 0.530 accuracy score compared to the vanilla version of Whisper which received an accuracy score of 0.485.

## Keywords

Vietnamese tones, Whisper, linear probing, speech representations, tonal languages, ASR interpretability

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## 1 Introduction

Vietnamese is a monosyllabic tonal language spoken by approximately 97 million people in both Vietnam and diaspora communities around the world. In Vietnam, there are two main dialects of Vietnamese: Northern Vietnamese and Southern Vietnamese. Northern Vietnamese (also called the Hanoi dialect) has 6 tones and is the prestige form of the language, used in academic and professional settings. The 6 tones in Northern Vietnamese are ngang, sắc, huyền, hỏi, ngã, and nặng. Southern Vietnamese (Saigon dialect) has 5 tones (hỏi and ngã have merged) as well as consonant mergers for certain consonants and consonant clusters. Southern Vietnamese also serves as the basis of many dialects in diaspora communities due to refugees from Southern Vietnam fleeing the war. Lexical tones in Vietnamese differ from one another in pitch contours and phonation. Lexical tones are

phonemically contrastive and change the meaning of the word depending on pronunciation. It is therefore of utmost importance that tonal information is considered and properly accounted for in respect to speech recognition systems.

Whisper is a multilingual speech recognition model developed by OpenAI [?] and released in 2022. It is an encoder-decoder transformer-based automatic speech recognition system (ASR) which was trained on approximately 680,000 hours of weakly-supervised transcribed audio scraped from the internet. It converts audio samples into a log-magnitude Mel-frequency cepstrum (log-mel spectrogram), which contains information on formants, pitch, timing, and intensity. This is then encoded, processed, and then passed through hidden layers for embedding. It has a standard transformer decoder which can be used for translation, speech production systems, and text generation.

In this study, we freeze a pre-trained model (Whisper) to study its internal representations of the Vietnamese language. This allows us to detect if information on lexical tones is truly being preserved and properly encoded. We train and use a linear classifier for probing, which is a simple interpretability technique allowing us to peek into the hidden layers of a model and analyze its internal representations.

The questions we attempt to answer in this paper are:

- (1) Are Whisper’s internal representations able to encode Vietnamese tonal information?
- (2) Does Vietnamese fine-tuning (such as PhoWhisper) improve lexical tone encoding?
- (3) Which hidden layer best encodes lexical tones and how does fine-tuning affect which layer best contains tone information?

## 2 Background

### 2.1 Vietnamese Tonal System

Table 1: Vietnamese Tone Contours and Phonation

| ID | Name  | Mark   | F0 Contour          | Phonation      |
|----|-------|--------|---------------------|----------------|
| 0  | Ngang | (none) | Mid level           | Modal          |
| 1  | Sắc   | á      | High rising         | Tense          |
| 2  | Huyền | à      | Low falling         | Breathy        |
| 3  | Hỏi   | ả      | Dipping             | Modal/tense    |
| 4  | Ngã   | ã      | Rising-glottalized  | Creaky         |
| 5  | Nặng  | ạ      | Low falling-stopped | Creaky/stopped |

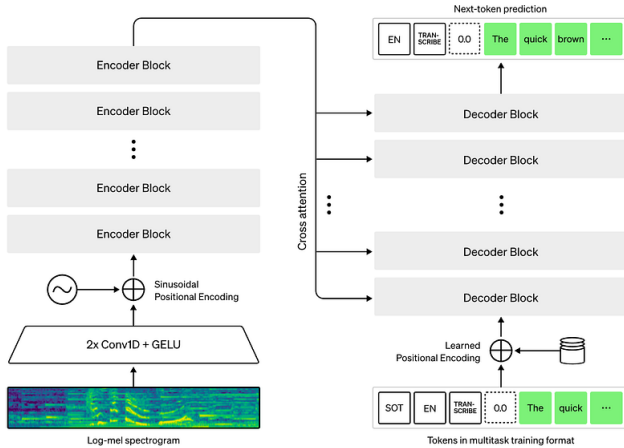


Figure 1: This diagram shows how sound is converted into an log-mel spectrogram before being passed into the encoder and decoder.

The tones of Vietnamese are described by their F0 contour phonation type. Ngang is the default way a vowel is pronounced, which can be described as flat or mid-level, produced with modal voice. Sắc is the high rising tone and is produced with tense voice. Huyền is the low-falling tone and is sometimes produced with breathy voice. Hỏi is the mid-falling tone which starts with modal voice, followed by tense voice. Ngã is the rising-glottalized tone which starts with modal voice, followed by creaky voice. Nặng is the low-falling stopped-tone, starting with tense voice until the vowel ends in a glottal stop.

Since Vietnamese tones carry lexical information, the definition of the word changes based on tone. A very common example is ma (ghost), mà (but), má (mother), mã (tomb), mã (code), and mạ (rice seedling). These minimal pairs show that tone is both lexically and phonemically contrastive.

## 2.2 Whisper’s Architecture

Whisper’s uses an encoder-decoder transformer as the base for its architecture and utilizes multi-headed self-attention. Input audio is sampled at a rate of 16,000 Hz and converted into a log-magnitude log-mel spectrogram. This is then passed into the encoder which processes and encodes information into hidden layers with pre-activation residual connections, producing an output with layer normalization. The residual connections allow the model to preserve information while continuing to learn from new transformations. Layer normalization helps stabilize the activation and controls mean and variance. This can then be passed into the decoder allowing the model to generate text, either for transcription, translation, or identification.

## 2.3 PhoWhisper

PhoWhisper is a Vietnamese fine-tuned version of OpenAI’s Whisper model [1]. It is designed to improve automatic speech

recognition performance for the Vietnamese language. It is created by fine-tuning Whisper on a 844-hour Vietnamese dataset allowing it to better encode Vietnamese pronunciation, vocabulary, and tonal information.

## 2.4 Probing Representations in Neural Networks

Probing classifiers are a simple, diagnostic tool used in interpreting a neural network’s hidden layers [2]. We use a simple logistic regression probe, which is a simple, lightweight model which is trained from the hidden layers on a frozen pre-trained model. If the probe is successful in predicting the target property (in this case tone), it suggests that said property was able to be encoded and the model has created an internal representation of the information.

## 3 Methodology

### 3.1 Data

The dataset we chose is fpt\_fosd from HuggingFace’s libraries. We originally wanted to use the FLEURS dataset developed by Google or the VIVOS dataset developed by the Vietnam National University, Ho Chi Minh City. We were unable to use the FLEURS dataset since it was not updated for the latest version of PyTorch, and VIVOS due to size constraints. The fpt\_fosd dataset was produced by the FPT Corporation in Vietnam and met our memory constraints. It is important to note that this dataset was prone to noise due to a lack of quality-control in recording. The dataset was already transcribed for us, allowing our probe to verify its predictions with the transcript.

### 3.2 Whisper Settings

For the model, we chose Whisper-small for the experiment, requiring significantly less virtual memory from the GPU and computational power. Its small size also allows for faster inference and hidden layer extraction, allowing us to quickly process a larger number of data samples. Despite being a lightweight version of Whisper, it still has the full encoder-decoder transformer architecture.

As for the model settings, we chose a sample rate of 16,000Hz, which is common in speech recognition applications due to it containing enough speech information from the audio as well as being computationally efficient. Vietnamese tones are primarily encoded in low-frequency pitch contours, which is well preserved at our chosen sample rate. It has 13 hidden layers, with the first being the embedding layer.

### 3.3 Probe Design

As for the probe, we chose a logistic regression (linear) probe with L2 regularization. This makes the probe prefer smaller, smoother weights and reduces noise from the dataset, as well as reducing overfitting and improving generalization. Due to the dataset having an imbalance of tones (since ngã is the rarest tone in the Vietnamese language), we chose balanced class weights. The probe is also designed to be trained independently per layer, and a 20/80 test/train split

was chosen. We left the regularization parameter as the default of  $C = 1.00$  to closely fit the data and keep the model simple. We also chose to seed the data, allowing us to control the randomness of the model. This lets us compare whether or not the model got lucky with the randomly chosen audio clips.

### 3.4 Experiment 1: Comparing Two Baselines

Our model was designed to parse the tones in an audio clip and return the most frequently heard tone in the clip. This was then compared with the transcript to measure accuracy. The model did not have access to the transcript during learning.

With the 20/80 test/train split, we first chose a sample size of 500 audio clips. This allows us to first train the data on 400 audio clips and then analyze the remaining 100. We first experimented with a linear probe to establish a baseline and general understanding of how our probe works. We tested 2 baselines to assess stability and ensure that similar results were being produced.

### 3.5 Experiment 2: Balanced or Unbalanced

The second method was to experiment with balanced class weights. Our method was to prioritize more marked features seen in Figure 2. This allows us to see if the dataset was a limitation, especially since Vietnamese has an imbalance of tone distribution in its lexicon.

|                                                       |              |
|-------------------------------------------------------|--------------|
| nặng > ngã > hỏi > sắc > huyền > ngang<br>most marked | least marked |
|-------------------------------------------------------|--------------|

Figure 2: Tiebreaker priority order for majority-vote tone labeling, ranked by acoustic markedness.

### 3.6 Experiment 3: MLP Probing

The third method was to try MLP probing. If MLP had better performance than our linear probe, that meant that tone information is non-linearly encoded in Whisper’s representations. This suggests that the tone information is encoded in a more complex way that requires a more powerful decoder to access.

### 3.7 Experiment 4: Random Seeding

Our fourth method was choosing different seeds. We used an initial seed of 42, followed by 7 and 13, all of which were arbitrarily chosen. If we got the same accuracy across different seeds, it shows that our probe is stable and the results were not artifacts of that set of data.

### 3.8 Experiment 5: Larger Sample Size

Our fifth method was to scale the probe up to 2000 samples from 500. The smaller, 500 sample dataset also allowed us to debug and adjust model parameters in a reasonable amount of time. With more data to learn from, we are able to better determine whether or not our results could be generalized

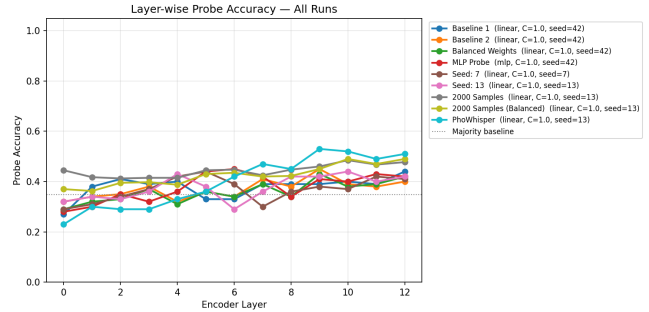


Figure 3: Accuracy with respect to layer. All experiments are shown in the legend.

beyond the smaller sample as well as whether or not the model can learn more intricate and robust representations.

### 3.9 Experiment 6: Fine-Tuned Model

Our final method was to test a Vietnamese fine-tuned model, PhoWhisper. This truly tells us whether or not fine-tuning the data improves lexical tonal representation within the Whisper model.

### 3.10 Baseline

We established two baselines to compare our results to:

- (1) Majority Class Baseline: 0.348. This is because out of 2000 samples, we got a majority of 696 ngang tones. This represents a classifier that always predicts the ngang tone regardless of audio input. A probe with this accuracy doesn’t learn anything and simply chooses the most likely probe every time.
- (2) Acoustic Baseline: 0.320. This baseline was processed manually from the same subset of data that the probe processes. It extracts 31 features, 5 from the fundamental frequency (F0) and 26 from the log-mel spectrogram. This represents what accuracy is possible from traditional speech processing before deep learning.

## 4 Results

### 4.1 Layer-wise Probing Accuracy

Figure 3 reports tone classification accuracy across encoder layers.

**4.1.1 Experiment 1: Comparing Two Baselines.** Our baseline experiment showed similar results for Baseline 1 and Baseline 2. This means that our probe is working properly. It also shows that the first probe had tone representation at L12 and the second had tone representation at L9. The discrepancy between Baseline 1 and Baseline 2 come from differences in class distributions and how well certain tones were represented and distinguished within the embeddings. Both runs used the same Whisper-small model, linear probing configuration, mean-pooling strategy and regularization settings.

**4.1.2 Experiment 2: Balanced or Unbalanced.** With the balanced class weights experiment, we can see prioritizing more marked tones can improve our accuracy. Since the ngang tone appears much more frequently than ngã and hỏi, we want to encourage our probe to learn the tone boundaries for all tones rather than optimizing for the most dominant tone. The balancing improves the probe’s sensitivity to those lesser represented tones.

**4.1.3 Experiment 3: MLP Probing.** In our MLP probing experiment, we can see that the accuracy remains the same as linear probing. This means that tonal representation within the hidden layers was able to be encoded and was not entangled in other confounding variables. This means that our linear probe was also working properly and that the task can be done with a simpler probing technique.

**4.1.4 Experiment 4: Random Seeding.** In our random seeding experiment, we get similar accuracy across all random seeds (testing 42, 7, and 13). This means that our model is working properly and that there weren’t any artifacts from subsets of the data that were chosen at random. This means that the probe did not get lucky with a certain dataset and outperform other runs.

**4.1.5 Experiment 5: Larger Sample Size.** In our larger sample size experiment, we can see how increasing the amount of samples improves the probe. We trade runtime and computational power for more accuracy, allowing the probe to learn from a more diverse set of data, helping improve tone boundaries. We can also see that the balancing improves our model even further, especially with a larger sample size. Our probe was able to better distinguish the tone boundaries between similar tones contours.

**4.1.6 Experiment 6: Fine-Tuned Model.** As expected, the PhoWhisper model proved to be the most accurate at 0.530. This fine-tuned version of Whisper noticeably improved the model’s performance. This suggests that tonal information is indeed being encoded and that the probe’s performance too benefits from fine-tuning. PhoWhisper’s additional exposure to Vietnamese speech reorganizes its internal representations in a way that makes tonal distinctions more linearly accessible to a probe.

## 4.2 Layer Analysis

We can generally break down the hidden layers into three main layers: early, middle, and later. The early layers encode low-level acoustic features, such as the shape of the log-mel spectrogram, basic phonetic features, etc. The middle layers encode intermediate representations, such as phonemes, tone contour, and voice quality. The later layers encode high-level linguistic features, such as abstract representations of the Vietnamese language. This would best be used for decoding tasks, we would ideally want to see our tonal representations earlier in the network.

**4.2.1 PhoWhisper: Best Layer Analysis.** In Figure 4, we are looking at the probe’s accuracy for the best run, using the

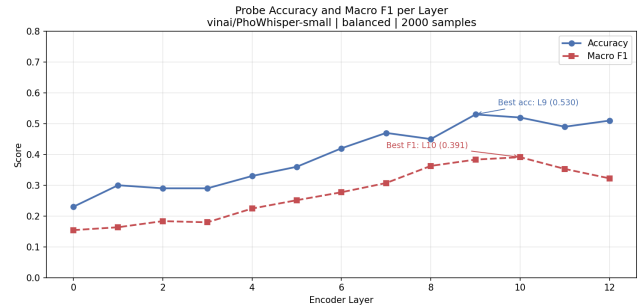


Figure 4: Above we can see the probe accuracy for the best-performing experiment with respect to hidden layers.

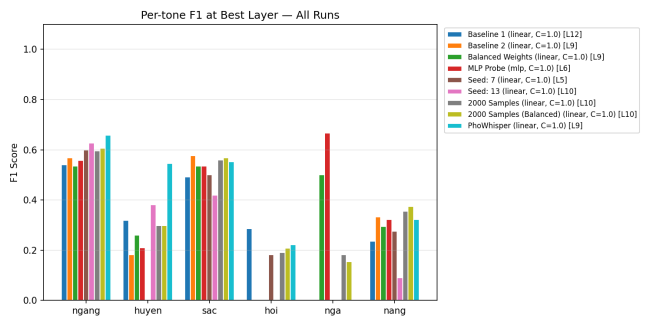


Figure 5: The F1 scores of all the tones can be seen per experiment.

PhoWhisper model. We can see that the accuracy surpasses both baselines around Layer 4, peaking at Layer 9. This suggests that the extra Vietnamese fine-tuning in training PhoWhisper caused tone information to consolidate in the middle-to-late layers, as opposed to the very last layer. This means that tonal information was being more efficiently encoded and its representations were being built earlier in the neural network, rather than the end which is prone to more confounding variables.

**4.2.2 Tone Distribution at Best Layer.** Figure 5 shows the F1 score distributed through each tone, as well as the experiments performed. We can see how the ngã tone is non-existent in all except for the MLP probe. However, we know that ngã is the rarest tone so that its high F1 score must be due to a statistical artifact. Since the MLP probe is more complex and has many more layers than a linear probe, this could cause it to memorize rare cases, such as in the ngã tone, and overfitting the data. It is also possible that it could be associating some incidental acoustic features in the recordings rather than the tone itself. Since our linear probe is much simpler, it is unable to overfit the data in such a way.

## 4.3 Confusion Analysis

**4.3.1 PhoWhisper: Confusion Analysis.** In the confusion matrix seen in Figure 6, we see that the ngã tone is being

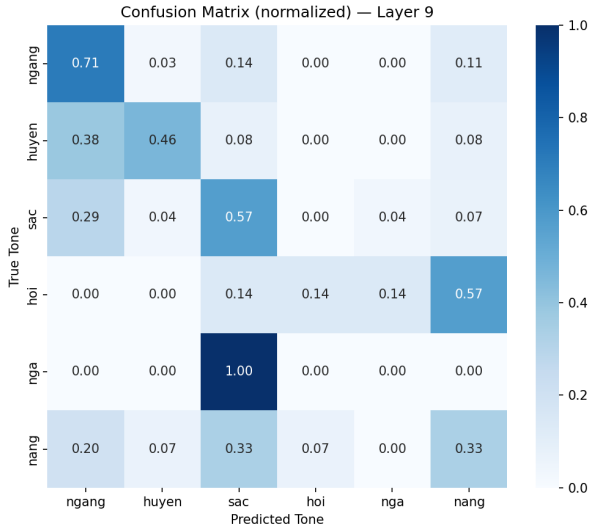


Figure 6: Confusion Matrix at Best Layer

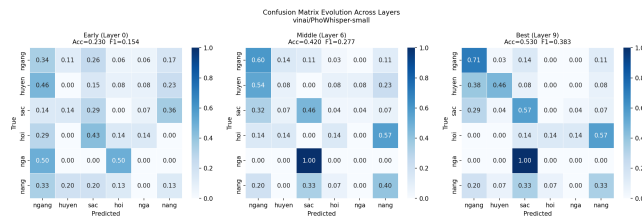


Figure 7: Confusion Matrix as it evolves through layers.

predicted as the *sắc* tone with a 1.00 accuracy. This is because the *ngã* tone is near non-existent in the dataset, with only 2 test samples. Both tones involve a high rising pitch, which could explain the confusion. Figure 7 shows the confusing matrix as it develops and gives us a glimpse in how the internal representations of the tones develop.

In the early layers, we see low accuracy for each tone. This means that the probe is barely working and suggests the earliest layers, especially at Layer 0, contain little to no tone information encoded. In the middle layers, we see improvement as the *ngang* tone rises to 0.60, *sắc* to 0.46, and *nặng* to 0.40. The *huyền* tone is still at 0.00, likely due to it being confused with the *nặng* tone as both are low. The *hỏi* tone is being predicted as the *nặng* tone at 0.57, meaning the probe is possibly picking up on the creaky phonation shared by both tones, but not the F0 contour patterns in the tones. In our best layer, Layer 9, we see that the *ngang* tone is at 0.71. It is well represented at this point in the neural network. *huyền* rises to 0.46, which is a huge improvement compared to earlier layers. This suggests that at this point, the probe is able to learn the contour differences between *huyền* and *nặng*. *sắc* reaches 0.57 and *hỏi* stays at 0.57, meaning it is still being confused for *nặng*.



Figure 8: The F1 scores for each tone can be seen per layer.



Figure 9: UMAP representation of all layers. Starred is the layer with the best accuracy.

Seen in Figure 8, we can see the F1 scores of all tones and how they develop throughout the layers. This suggests internal representations of lexical tones being built throughout the network layers.

#### 4.4 UMAP Projections

Figure 9 shows us the UMAP projections of the PhoWhisper-small encoder at each layer. As seen in Figure 10, the representations are scattered all over the plot. At the very first layer, the probe did not learn any tonal information and is unable to differentiate them. As we approach Layer 6, we see partial organization, especially with the *ngang* and *sắc* tones starting to separate from the other tones. At our most accurate layer, Layer 9, we see the greatest degree of tonal representation. *Hỏi*, and *nặng* show heavy overlap, owing to the confusion observed in the confusion matrix seen in Figure 6.

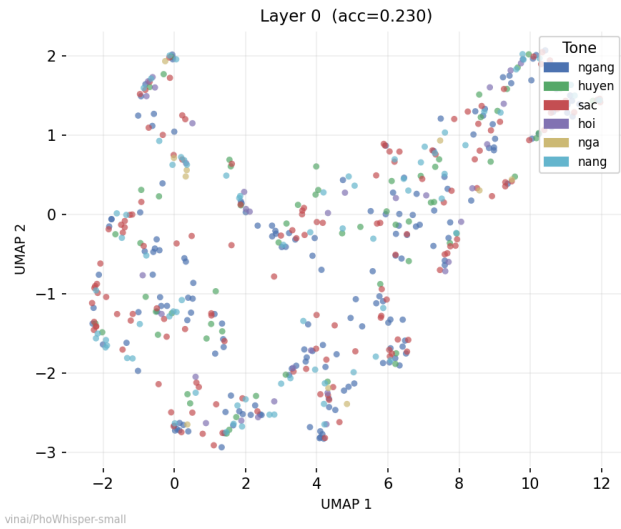


Figure 10: UMAP representation of Layer 0. As we can see the tones are scattered all around and are not organized.

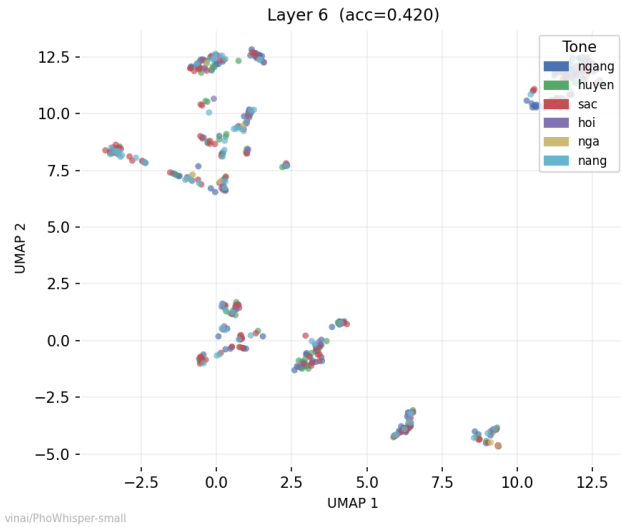


Figure 11: UMAP representation of Layer 6. We can see that the tones are starting to group together and form clusters.

## 5 Discussion

### 5.1 Layer Peak Shift

For the vanilla version of Whisper, we can see how the accuracy peaks at Layer 12. This suggests that tone information is formed at the end of the network. However, for PhoWhisper, we can see it peaking at Layer 9, showing that fine-tuning can cause tone representations to show up earlier in the neural network. This means phonological information such as lexical tones is more accessible at the intermediate processing stages of the model.

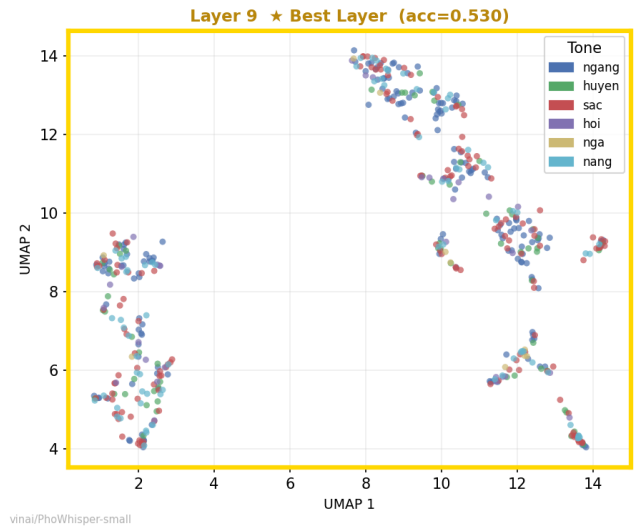


Figure 12: UMAP representation of Layer 9. This was the most accurate layer and gives us the best picture of how tones were represented in model.

### 5.2 Huyền Phase Transition

Most noticeably, we can see that the huyền tone showed the most improvement, from an F1 of 0.00 at Layer 6 to an F1 of 0.546 at Layer 9. This tells us that the huyền tone may be more complex to process, requiring higher level processing to encode. It may be due to the breathy, low-falling phonation that the huyền tone possesses, distinguishing it from other tones. PhoWhisper was able to encode this information, however vanilla Whisper struggled to do the same, showing that the Vietnamese fine-tuning did indeed improve its sensitivity to phonation types and in turn, help it distinguish and encode more complex tone information.

### 5.3 Hỏi-Nặng Confusion

We see in Figure 5 that the hỏi tone is consistently being predicted as nặng across all layers. This may be due to both tones sharing creaky phonation, which is distinguished in the F0 contour. This may also point to mean-pooling affecting the contour shapes of the tones, losing much of the finer information that could help distinguish these two tones. In future studies, we could use syllable-level alignment using the Montreal Forced Aligner to annotate our audio. This is a much more involved study as it requires significant work in preparing the data due to the large sample size required to train the model.

### 5.4 Logistic Regression vs MLP

Our results showed that our MLP probe did not outperform the logistic regression (linear) probe. This tells us that tone information in Whisper's representations are linearly accessible and do not require any complex transformations to decode.



## 5.5 UMAP and Tone Representations

The UMAP representations seen in Figure 9 and Figure 12 show us how internal representations of lexical tones in PhoWhisper are gradual and incomplete. Even after projection to two-dimensions, they show considerable overlap. However, this is to be expected since for each audio clip, there are multiple tones present. Our model only returns the most frequently occurring tone in each sample. Mean pooling across all the clips when converting audio to MCP may also cause the model to produce an representation of the average of all the tones in each clip, rather than each individual tone. Despite this overlap, we can still organization and some representation.

## 6 Conclusion

The linear probing of Whisper and PhoWhisper showed that tonal representation was indeed possible to encode. We can

see that fine-tuning a model specifically to Vietnamese significantly improved the accuracy compared to a general, multi-lingual model, with PhoWhisper achieving 53% accuracy and Whisper only achieving 48.5% accuracy. This fine-tuning also caused a shift in the huyền tone, encoding its information earlier in the neural network. In sum, fine-tuned models are able to encode complex lexical tones showing results in even smaller models with a limited dataset.

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